

DHANANJAYAN R

Python Developer

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SUMMARY

Backend Engineer / Product Engineer with strong experience building scalable REST APIs and high-throughput processing systems using Python, Flask, and FastAPI. Proven track record optimizing Apache NiFi pipelines and developing custom Python ExecuteScript processors for XML/JSON/EDI transformations, scaling logistics workloads to a stable ~2,000 TPM under constrained resources. Extensive hands-on expertise in distributed architectures using RabbitMQ, Redis, PostgreSQL/MySQL, Docker, Nginx, AWS (S3/EC2), and Linux, along with server resource tuning and Grafana-based observability. Strong in performance engineering (CPU, memory, disk IOPS, DB throttling), queue prioritization, and production troubleshooting, with consistent cross-team collaboration and ownership. Experienced in operationalizing ML/NLP/NER outputs as APIs (implementation-focused) to ensure reliability and scalability in real environments. Earlier experience includes JavaScript-driven dashboards (AJAX, jQuery, vanilla JS, HTML/CSS/Bootstrap), web scraping (BeautifulSoup, Selenium), and data processing with Pandas and OpenCV.

Experience: 4.5+ years (Full-time Backend/Product Engineering) + 5 months (Internship)

EXPERIENCE

Member of Technical Staff – Product Engineering

26/03/2025 – Present

(Python Backend / Performance), EdgeVerve (Infosys) ✉

Hybrid (Bangalore)

- Owned and Implemented performance optimization for **Apache NiFi pipelines** supporting **logistics EDI** workflows; improved stability and scaled throughput to **~2,600 TPM** under constrained infrastructure by Delivering architecture and scaling recommendations in Process.
- Built and maintained **custom Python NiFi processors** to extend platform capabilities, enabling complex **XML ↔ JSON ↔ EDI** transformations and faster integration delivery.
- Optimized pipeline performance using **native NiFi capabilities** (processor tuning, back-pressure, concurrency, batching, scheduling, Queue Prioritization) and reduced overhead in **custom API calls**.
- Strengthened reliability through **observability improvements**: upgraded/tuned **Grafana**, implemented monitoring dashboards, and automated event-level and summary-level metrics reporting.
- Diagnosed production bottlenecks across **CPU usage, memory utilization, disk IOPS, and DB server throttling**.
- Designed/implemented a **rule engine** for dynamic JSON manipulation based on business requirements, enabling configurable transformations without repeated code changes.
- Led cross-team debugging and support, translated business requirements into technical design, and accelerated delivery by enabling reusable Python components and automation.

Tech: Python, Apache NiFi, ExecuteScript, XML/JSON/EDI, RabbitMQ/Redis, PostgreSQL, Linux, AWS, Exposure to Kubernetes & Grafana.

Software Developer (python), TNQ Technologies

01/02/2023 –

21/02/2025

Chennai

- Developed and maintained **REST API-based applications** using **Python**, integrating **Flask, RabbitMQ, Redis, SQL, Docker, Nginx**, and API documentation via **Swagger/OpenAPI**. (Smart Language)
- Implemented queue-driven workflows and domain-specific automation, improving reliability and throughput using **asynchronous processing** and structured pipeline stages.
- Delivered a phased extraction solution for complex documents Citation Project (Smart Citation With NER integration) : **HTML element extraction, pattern matching**, and automated **XML tag addition**, ensuring quality and consistency.
- Contributed to ML implementation initiatives such as **NER integration**, focusing on operationalization and pipeline readiness; used **MLflow** for experiment tracking / MLOps concepts.
- **2nd Runner-up (GENAI Hackathon)**: Built an AI-based smart configuration solution, recognized for practical implementation and impact.

Tech: Python, Flask, REST APIs, RabbitMQ, Redis, SQL, Docker, Nginx, Swagger/OpenAPI, AJAX, JavaScript, Linux

Junior software programmer (Python), Genome International India Pvt Ltd

01/06/2021 –

31/01/2023

Chennai

- Built backend services and middleware using **Python (Flask)**; supported frontend requirements with **JavaScript**, and worked alongside legacy components (PHP).
- Helped modernize architecture by moving existing services into **Flask-based REST middleware**, improving modularity and maintainability.
- Worked on data manipulation and reporting using **Pandas**, created dynamic dashboards using **HTML/CSS** and **JavaScript charting (AMCharts)**.
- Implemented web scraping pipelines using **BeautifulSoup** and **Selenium WebDriver** for gene cards and research articles, supporting curated dataset creation.
- Applied **biocuration workflows** and assisted in **NLP-based entity detection**, contributing to biomedical dataset enrichment and quality improvements.
- Supported deployments and automation via **AWS EC2, Linux**, and **Shell scripting**, assisting with production rollouts and operational scripts.

Tech: Python, Flask, JavaScript, HTML/CSS, AMCharts, Pandas, BeautifulSoup, Selenium, AWS EC2, Linux, Shell scripting

Promoted from: Software Developer Intern (Python)

15/01/2021 – 31/05/2021

- Assisted in building backend utilities and automation scripts; contributed to scraping and data curation tasks under R&D team guidance.

EDUCATION

Bachelor of Engineering (B.E.), Mechanical Engineering, Annamalai University

2015 – 2019

GPA: 8.65/10

CAPSTONE PROJECT

Experimental Studies on Solar Still (Water Purification)

- Designed and executed experiments to optimize solar still efficiency for household water purification

SKILLS

Backend / API Engineering

Python, REST APIs, Microservices, Flask, FastAPI, ORM (SQLAlchemy/ORM concepts), OpenAPI/Swagger, Postman

Cloud / DevOps / Platform

AWS (S3, EC2), Docker, Nginx, Linux/Ubuntu, Git/GitHub, CI/CD pipelines, observability/monitoring (Grafana)

Web / UI (Supporting)

JavaScript, AJAX, jQuery, HTML, CSS, Bootstrap

Exposure

MLflow, spaCy (NER), SciSpacy, Matplotlib, Seaborn, AMCharts, Boto3, Hugging Face Transformers, LangChain, SERRANT, NLTK

Distributed Systems & Async Processing

Async processing, message-driven architecture, RabbitMQ, Redis, Apache NiFi (ExecuteScript), queue prioritization, throughput tuning, Redis caching, rate limiting

Performance & Testing

Performance engineering (CPU, memory, disk IOPS, DB throttling), load testing, profiling and bottleneck analysis

Scraping / Extraction

BeautifulSoup, Selenium WebDriver

Databases

PostgreSQL, SQL, connection pooling (PgBouncer)

Data Processing & Automation

XML, JSON, EDI transformations, Pandas, NumPy, Advanced Regex, OpenCV, OCR (Tesseract)

Tools

Visual Studio Code, Jupyter Notebook, JIRA, Windows OS, Linux OS

TRAINING/COURSES

- The data science course by 365 careers [↗](#)
- Python for data science and machine learning [↗](#)
- The Complete SQL Boot camp [↗](#)

PROJECTS

Colorizer, <https://github.com/dhananjayan-r/Colorizer> [↗](#)

- Built an interactive **image colorization** web app that converts grayscale images into plausible color outputs using an OpenCV-based processing pipeline.
- Packaged the solution as a lightweight **Streamlit** application with a simple UI for image upload, processing, and preview.
- Focused on **implementation and user-facing delivery** (app workflow, image handling, performance, and stability).

Tech: Python, OpenCV, Streamlit

LinkedIn Scraper, https://github.com/dhananjayan-r/Linkedin_scraper [↗](#)

- Developed a Python automation script to scrape LinkedIn job listings within a selected timeframe (e.g., yesterday / last month).
- Extracted structured job details (job description, hiring manager name, company, and related fields) using **Selenium + BeautifulSoup**, and designed storage for persistence in a database.
- Created a reusable scraping workflow with modular extraction logic to support evolving page structures and selectors.

Tech: Python, Selenium WebDriver, BeautifulSoup

Swagger API, <https://github.com/dhananjayan-r/Swagger-open-api> [↗](#)

- Built a **spec-first REST API** where endpoints are defined in an **OpenAPI (YAML)** contract and become runnable with minimal boilerplate.

Tech: Python, OpenAPI/Swagger (YAML), Connexion